



Interface-Conditioned Energetic Descriptions

Diagnostic discipline note (standalone; non-prescriptive)

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Adjacency (non-derivative / non-prerequisite): This note is structurally adjacent to other PortusSophia™ research artifacts. It may be cited as a companion discipline note (e.g., alongside "Constraint-Dynamics") without establishing a prerequisite relationship.

Posture: Diagnostic; non-prescriptive; standalone.

Abstract

This note proposes a minimal, non-agentic formalization of interface conditioning: the transformation of a description under an explicit interface declaration (e.g., coarse-graining, projection, restriction, partition, or constraint-preserving mapping). The scope is intentionally descriptive: the interface is treated as a declared boundary condition on legibility, not as an agent, mechanism, or causal operator. The paper introduces vestigium as the persistent remainder under interface conditioning—residual constraint, compressed correlation, or untranslated structure that does not become legible within the declared interface. "Descriptive collapse" is used strictly as a representational term for interface-conditioned resolution into a finite descriptive mode; it is not advanced as an ontic disclosure or epistemic privilege claim. The central claim is limited: descriptions are interface-relative, non-equivalent across interfaces, and commonly non-invertible under reduction, with residue persisting as an auditable remainder of the chosen interface.

A scope-delimited inquiry within the PortusSophia™ research sequence.

Section 0. Discipline and Explicit Exclusions

This paper operates under strict descriptive discipline. It does not advance claims about what exists, why phenomena occur, or how they ought to be interpreted or acted upon. The following exclusions are explicit and non-negotiable:

- No agents, observers, subjects, or decision-making entities are introduced.
- No signaling theory is invoked.
- No teleology, optimization objective, or "purpose" language is assumed.
- No ethical prescriptions, governance proposals, or "best practices" are provided.
- No ontological claims are made about what energy "is," what the world "really is," or what entities "truly exist."
- No mechanism claims are advanced (no causal story of how one substrate produces another).

The paper instead describes a constraint-conditioned relationship between: (i) an explicit interface declaration, (ii) what becomes legible within that interface, and (iii) what remains as residue under that interface.

Section 1. Orientation

Many cross-domain debates degrade into ambiguity because the descriptive interface is left implicit. The same interaction may be described as "force," "tension," "vibration," "field," "information," "loss," or "trauma," depending on the interface imposed.

This note formalizes a minimal descriptive stance:

- Interfaces condition descriptions.
- Descriptions are interface-relative.
- Residue persists under interface conditioning.

Section 2. Interface Declaration

An interface is defined here as an explicit constraint declaration that determines what counts as legible within a description.

An interface may be expressed as:

- A coarse-graining map
- A projection or restriction
- A partition or equivalence relation
- A measurement basis or observable set (treated descriptively, not ontologically)
- A loss function or scoring rule (treated descriptively, not teleologically)

The interface is not an agent and does not "act." It is a declared boundary condition on description.

Section 3. Energetic Descriptions as Interface-Relative Collapses

"Energy" is used in a deliberately conservative way: as a placeholder for cross-domain descriptions of persistence, capacity-to-affect, or constraint-bearing interaction.

Within an interface, energetic descriptions appear as local descriptive forms:

- Vibration as an oscillatory legibility mode
- Force as directionalized constraint legibility
- Tension as interface-bounded persistence of relational constraint

These are treated as interface-conditioned collapses of a more general interaction that is not claimed as ontological substrate.

Section 4. Vestigium (Residual Constraint)

Define vestigium as the persistence of constraint-conditioned remainder under interface conditioning.

Vestigium is not treated as error, pathology, or moral deficit. It is the residual structure that remains when an interface is applied.

Vestigium may be:

- Unrendered degrees of freedom
- Compressed or discarded correlation
- Untranslated structure under mapping
- Residual cost displaced outside the declared interface

Vestigium can be described, but it cannot be eliminated by definitional fiat.

Section 5. Descriptive Collapse

Descriptive collapse names the resolution of pre-descriptive interaction into a finite descriptive mode under constraint. The term is used strictly in a representational sense and does not denote a physical process or causal event. Interface-conditioned resolution is a legibility event relative to a declared boundary, not an ontic disclosure and not an epistemic privilege claim.

The operative claim is limited:

- Under a declared interface, only certain descriptive invariants remain legible.
- The remainder persists as vestigium.

Section 6. Non-Equivalence Across Interfaces

Interface changes do not preserve equivalence of descriptions.

A description that is coherent under one interface may be incoherent under another without implying contradiction. The non-equivalence is structural: the interface changes what counts as a stable descriptive form.

This paper therefore treats translation across interfaces as a bounded operation that may fail.

Section 7. Irreversibility and Loss of Degrees of Freedom

When an interface reduces degrees of freedom (by projection, coarse-graining, or restriction), the reduction is generally non-invertible.

This is not asserted as a metaphysical claim; it is a structural property of many interface operations.

The consequence is operational:

- Some structure becomes non-recoverable under the declared interface.
- That non-recoverability is not "error"; it is the interface's cost.

Section 8. Failure Modes (Hostile-Reading Safety)

The following drift risks are explicitly bracketed:

1. **Observer smuggling:** treating the interface declaration as a hidden agent.
2. **Ontology inflation:** treating descriptive forms (force/vibration/tension) as claims about substrate.
3. **Totalization:** treating interface-relative descriptions as universal.
4. **Moralization:** treating vestigium as guilt, defect, or obligation.

These are failure modes of reading, not conclusions of the framework.

Section 9. Compatibility and Adjacency

This descriptive layer remains compatible with other conceptual domains—including entanglement, awareness, consciousness, reason, and ethics—without governing or subsuming them. Compatibility does not imply dependence, priority, or derivation. In applied contexts, it may be cited as a companion discipline note (e.g., alongside Constraint-Dynamics) without establishing a prerequisite relationship.

The present note remains strictly pre-agentic and pre-ethical.

Section 10. Closing Scope Statement

This paper describes how interfaces condition energetic descriptions and how vestigium persists under that conditioning.

It does not claim:

- a unified physical theory,
- a metaphysical substrate,
- a moral program,
- a governance proposal,
- or a mechanism of emergence.

It is a discipline note: a structural constraint on interpretation.

Appendix A. Worked Exemplar (Interface Declaration Only): Double-Slit as Interface Non-Equivalence

This exemplar is included solely to show how an explicit interface declaration changes what counts as legible. It makes no claim about physical mechanism, observer causality, or ontological substrate. The double-slit setup is treated as a well-known case where different measurement interfaces yield non-equivalent descriptive forms.

Interface A (Interference-legibility): Declare an interface in which the admissible description is the distribution of impacts on a screen without which-path resolution. Under this interface, the stable descriptive form is an interference pattern: correlation structure is legible only at the level of aggregate distribution.

Interface B (Which-path-legibility): Declare an interface in which the admissible description includes which-path resolution (path distinguishability). Under this interface, the stable descriptive form is path-indexed detection statistics; the interference pattern is not preserved as a legible invariant under the declared constraints.

Non-equivalence statement (IC-local): The two interfaces do not preserve descriptive equivalence. A description coherent under Interface A may be incoherent under Interface B without implying contradiction. The shift is treated as an interface-conditioned change in what is legible, accompanied by remainder (vestigium) relative to whichever interface is applied.

Vestigium note: Under Interface A, path-resolved structure is not rendered as a legible invariant; under Interface B, interference-structure is not rendered as a legible invariant. This note treats those losses of legibility as interface-relative remainder, not as error, defect, or ontic elimination.

Provenance and Responsibility

This document is authored under a boundary-discipline posture: terms are used as structural operators rather than ontological claims, and no prescriptions or governance proposals are advanced. Any computational tools used in drafting are treated as non-authoritative aids; responsibility for content selection, scope, and meaning remains with the human author.

References (Contextual; Non-grounding)

Purpose: These items are supplied to reduce vocabulary-collision and expectation mismatch. They are not invoked as proof, mechanism, or authority. Citations function as contextual pointers only and do not establish derivation or prerequisites.

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Companion discipline note (local citation)

- Dennis, J. R. (2026). *Interface-Conditioned Energetic Descriptions* (companion discipline note), v0.1.1.

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